

HELMCKEN ISLAND, BC, Canada

WMO: 714813

Lat: 50.400N

Lon: 125.870W

Elev: 19

StdP: 101.10

Time Zone: -8.00 (NAP)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.0	-13.0	1.2	-1.2	-9.8	1.6	-0.8	16.4	6.9	15.8	6.7	4.7	90	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	21.7	14.1	20.2	13.6	19.0	13.2	15.5	19.4	14.8	18.2	14.2	17.3	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.1	16.0	13.4	9.6	15.3	12.8	9.2	14.9	43.4	19.2	41.4	18.4	39.7	17.6	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.8	10.9	DB	-4.6	26.3	2.6	1.7	-6.5	27.5	-8.1	28.5	-9.5	29.5	-11.5	30.7	(4)
(5)			WB	-6.6	16.8	2.6	1.4	-8.6	17.8	-10.1	18.6	-11.6	19.4	-13.5	20.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	3.8	5.0	6.3	8.7	11.4	13.2	15.3	15.4	13.2	9.2	6.0	3.9		
	DBStd	4.68	2.83	2.26	2.03	1.98	2.27	1.87	1.59	1.75	1.68	1.99	2.38	2.76		
	HDD10.0	856	192	141	116	49	11	0	0	0	1	38	119	189		
	HDD18.3	3293	450	374	373	290	215	153	93	93	153	283	368	448		
	CDD10.0	607	0	0	1	9	54	98	166	167	98	14	1	0		
	CDD18.3	2	0	0	0	0	0	0	1	2	0	0	0	0		
	CDH23.3	17	0	0	0	0	2	1	6	8	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0		
(14) Wind	WSAvg	3.7	4.5	4.5	4.1	3.8	3.2	2.9	2.8	2.6	2.7	3.7	4.5	5.2		
(15) Precipitation	PrecAvg	2091	274	166	170	147	101	92	64	74	106	254	320	300		
	PrecMax	2598	435	427	305	333	215	192	129	166	275	391	514	440		
	PrecMin	1578	141	2	17	41	42	38	8	5	20	53	141	150		
	PrecStd	264	85	93	65	80	41	41	35	48	58	102	106	71		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	10.9	13.6	17.7	22.0	22.5	23.8	24.3	20.5	16.1	11.6	9.4		
		MCWB	8.1	7.5	8.4	10.1	12.4	14.1	15.2	15.3	13.5	12.3	10.2	7.0		
	2%	DB	8.6	9.4	11.2	14.8	18.7	19.9	21.4	21.5	18.6	13.8	10.5	8.2		
		MCWB	7.2	7.1	7.2	9.0	11.7	13.1	14.3	14.4	13.1	10.5	8.9	6.4		
	5%	DB	7.8	8.6	10.1	13.0	16.6	18.1	19.9	20.0	17.2	12.7	9.7	7.7		
		MCWB	6.5	6.7	6.8	8.1	10.9	12.4	13.7	13.8	12.6	10.0	8.1	6.1		
	10%	DB	7.2	7.9	9.2	11.7	15.1	16.6	18.7	18.7	16.1	11.7	8.9	7.0		
		MCWB	6.1	6.3	6.8	7.9	10.1	11.5	13.1	13.4	12.1	9.5	7.5	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	9.1	9.5	11.1	13.8	15.2	16.4	17.2	15.5	13.2	10.8	8.1		
		MCDB	9.3	10.2	11.5	14.6	18.6	20.6	22.2	21.0	17.7	14.7	11.6	8.8		
	2%	WB	7.5	8.0	8.7	10.3	12.7	14.0	15.2	15.7	14.4	11.5	9.3	6.9		
		MCDB	8.3	8.6	10.0	12.9	16.8	17.9	19.6	19.0	16.7	13.0	10.3	7.7		
	5%	WB	6.9	7.3	8.1	9.6	11.9	13.3	14.6	15.0	13.7	10.6	8.5	6.3		
		MCDB	7.6	8.2	9.4	11.8	15.1	16.3	18.2	18.2	15.9	12.1	9.5	7.2		
	10%	WB	6.3	6.6	7.5	8.8	11.1	12.5	13.9	14.3	13.0	10.0	7.8	5.8		
		MCDB	7.1	7.7	8.8	10.8	14.2	15.3	17.1	17.3	15.1	11.4	8.8	6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.6	3.6	4.6	5.6	6.5	6.0	6.7	6.7	5.8	3.7	2.7	2.5		
		MCDBR	2.9	4.3	6.1	8.2	9.6	9.0	8.9	9.1	7.8	5.4	3.2	3.0		
	5% WB	MCWBR	2.3	2.6	2.9	3.4	3.9	3.6	3.6	3.6	3.6	3.0	2.4	2.2		
		MCWBR	2.7	3.2	4.5	6.9	8.3	7.9	8.0	8.2	6.8	4.7	2.9	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.316	0.343	0.362	0.379	0.377	0.364	0.367	0.342	0.366	0.321	0.308		
		taud	2.403	2.476	2.418	2.376	2.371	2.415	2.492	2.497	2.557	2.380	2.457	2.370		
	Ebn at Noon		722	828	862	882	878	880	888	869	858	746	708	661		
		Edh at Noon	63	75	94	110	115	111	102	96	82	82	60	56		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.60	2.40	3.76	4.94	5.25	5.72	4.96	3.42	1.83	0.88	0.60		
	RadStd		0.12	0.29	0.35	0.39	0.55	0.59	0.68	0.42	0.32	0.33	0.17	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page